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CYB102 | Intermediate Cybersecurity

Intermediate Cybersecurity Fall 2024 (@ Section 1 | Tuesdays 5PM - 7PM PDT)

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Need help? Post on our **class slack channel** or email us at **support@codepath.org**

- Getting Started
- Course Overview
- Schedule
- VM Access Guide
- AI Usage
- My Report
- Individual Projects
  - Unit 1
  - Unit 2
  - Unit 3
  - [redacted]
  - Unit 5
  - Unit 6
  - Unit 7
- Group Capstone
- Groupings
- Career Resources
- Prework
- Survey
- Overview
- Lab
- Project
- Resources
- Career

Lab #4: Monkey in the Middle

🏁 **MACHINE ACCESS CHECK** 🏁

To proceed with this unit, please ensure you have access to a ***cyb102 Ubuntu VM*** as detailed in the 🔗 **VM Access Guide**.

We recommend you begin with **Azure Labs**.

✅ **All options** in the 🔗 **VM Access Guide** will work for this unit.

Overview

In today's lab you will first capture encrypted HTTPS traffic in Wireshark. Then you'll install mitmproxy and view the same encypted traffic, but through a proxy server.

After determining the similarities and differences in the information captured, you'll gain some practice intercepting network requests and modifying them using a proxy server.

🎯 Goals

By the end of this lab you will be able to...

Analyze network traffic using Wireshark
Install and configure mitmproxy on your virtual machine
Analyze network traffic using mitmproxy
Modify intercepted network requests

Note: There are no stretch goals for this lab.

Resources

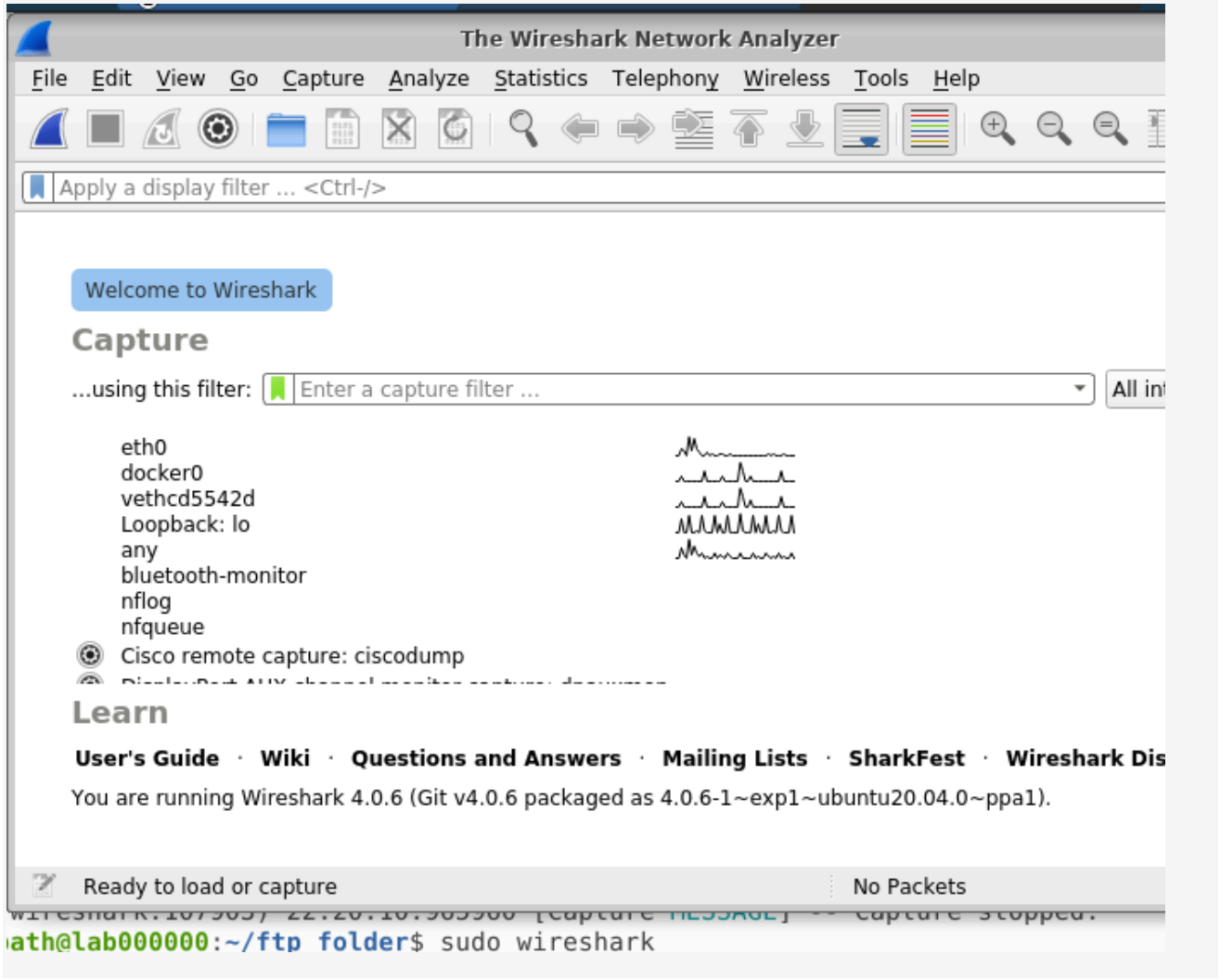
HTTP Response Status Codes
mitmproxy Tutorial
mitmproxy Docker Install mitmproxy via dockerhub

Lab Instructions

👥 Exercise 0: Analyze Network Traffic with Wireshark

Start up Wireshark and visit any encrypted website (ex: https://www.codepath.org/)
If you are using the Azure VM lab, run `sudo wireshark` from the terminal and select the network interface `eth0`

Click here for 📸 Example Screenshot of using Wireshark



Take some time to look through the packets you captured, and what information is available to you

👥 Exercise 1: Analyze Network Traffic with mitmproxy

Step 0: Install mitmproxy

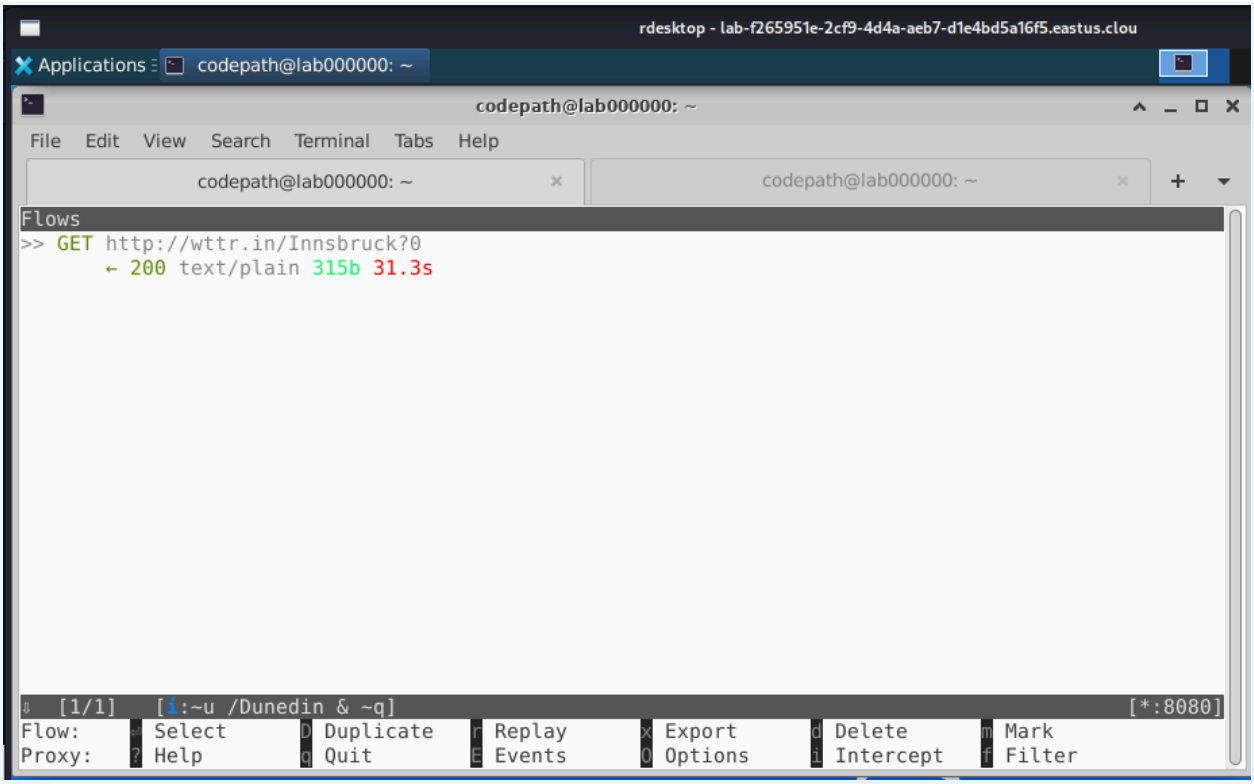
Mitmproxy isn't pre-installed on our machine, so we'll want to complete these steps to make sure it's up and running as intended.

> You should have `docker` installed from Unit 3. If not go ahead and run `sudo apt install docker.io`
> With docker installed run the following on your virtual machine:

```
sudo docker run --rm -it -v ~/.mitmproxy:/home/mitmproxy/.mitmproxy -p 8080:8080 mitmproxy/mitmproxy
```

> `mitmproxy` should be running after this one-liner and open

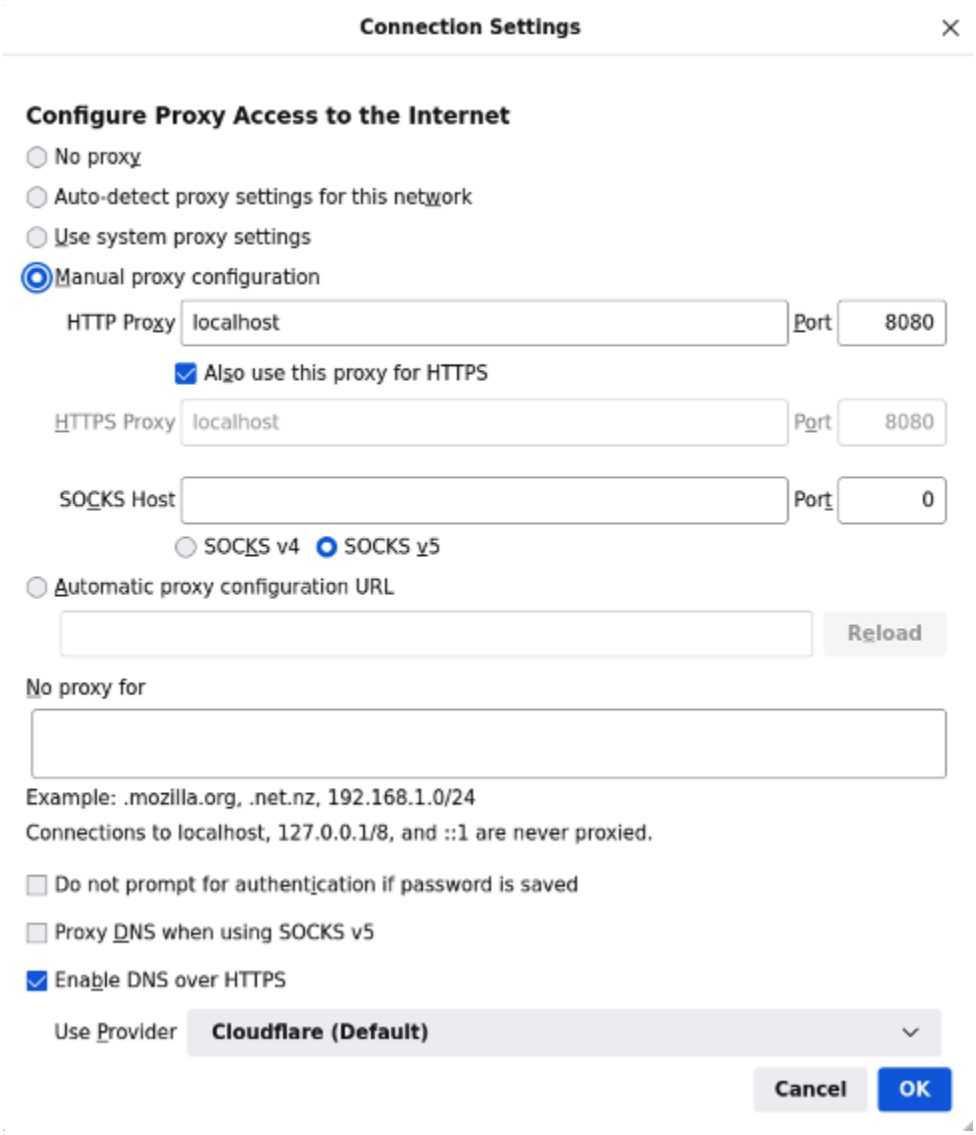
❓ What does mitmproxy look like?



🔥 Are there other ways to install mitmproxy

Step 1: Configure Network Settings

Run `firefox` to open the browser, and click on the hamburger icon in the top right to open the settings
Scroll down to the network settings, and configure Firefox to allow a proxy

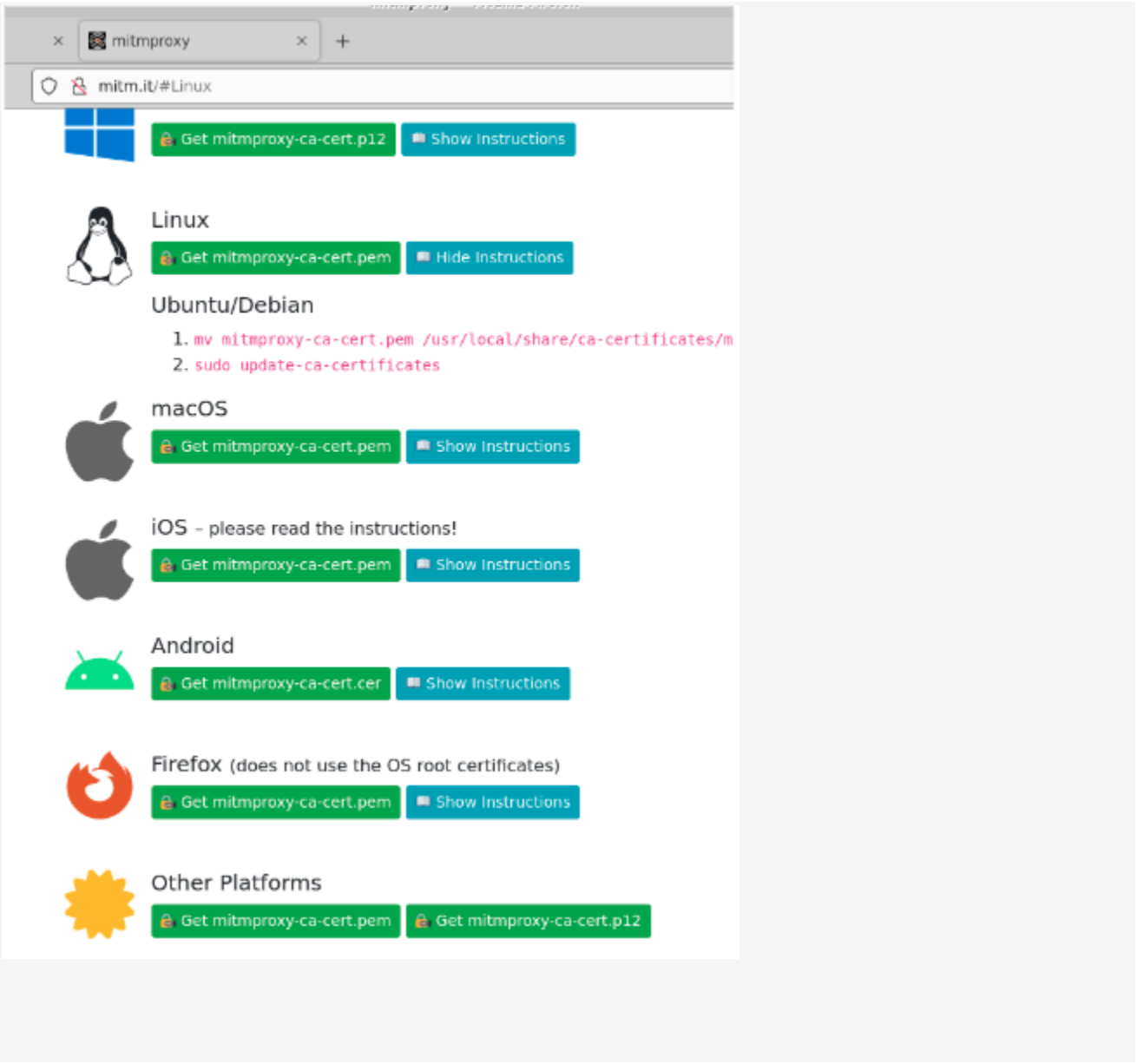


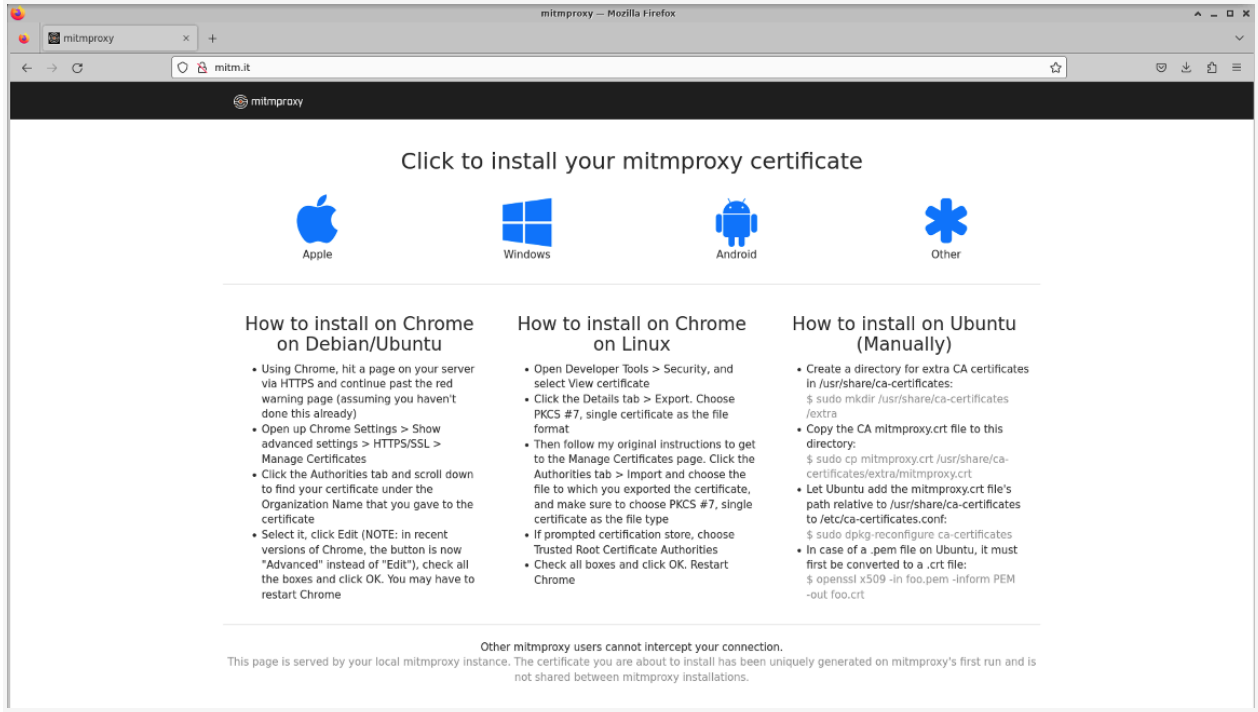
In your Firefox browser, go to http://mitm.it
Select the "Linux", "Firefox", or "Other" option (in order of preference), and follow the installation instructions.

- 💡 Tip: Don't see instructions? Make sure `mitmproxy` is running!

Note: Especially make sure to complete the last step of converting the `.pem` file to a `.crt` file

Click here for 📸 Example Screenshots of Installation Instructions





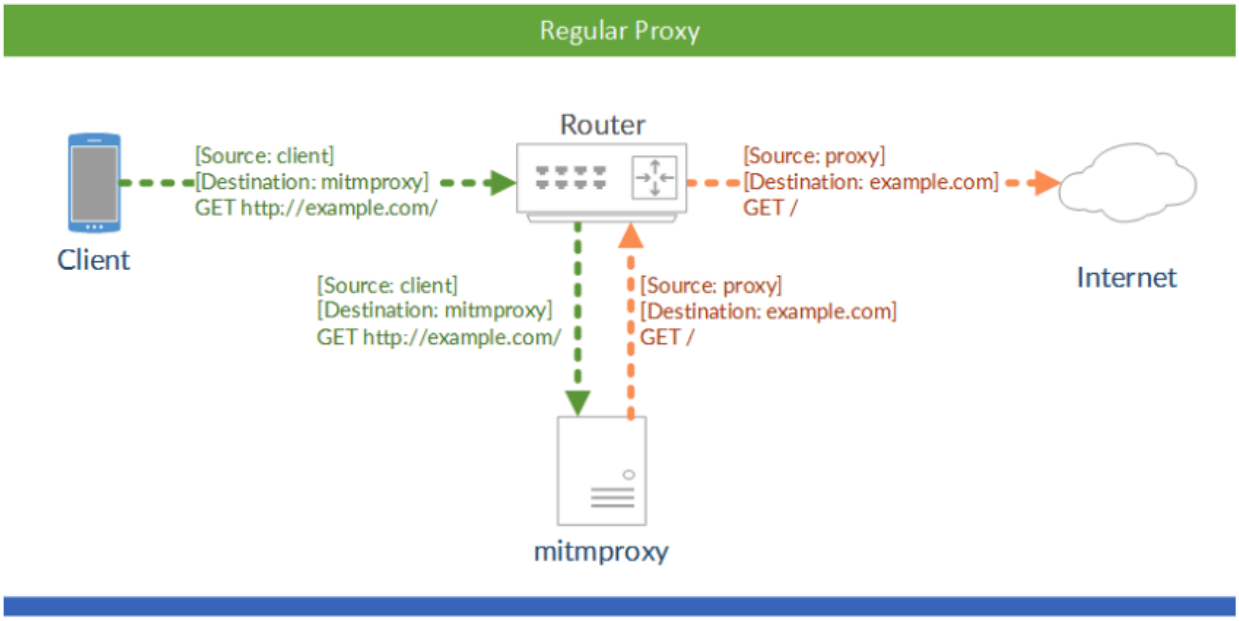
Now that we've downloaded the certificate, you can add it to Firefox so it trusts you to browse the web through the mitmproxy server

Tip: The certificate should be located in your `~/Downloads` folder

Step 2: Start Analyzing with mitmproxy

With `mitmproxy` running, revisit the same encrypted website you went to earlier (ex: https://www.codepath.org/)
We should see logged requests similar to capturing packets on Wireshark
Select one of the requests, and a more detailed view should open
Take some time to explore the information offered, and think about how it differs from what Wireshark provides

- Is there more or less detailed information?
- Given that this is an encrypted website (HTTPS), why can you see more details using mitmproxy?



👥 Exercise 2: Modify Requests with mitmproxy

Now we'll be following the mitmproxy tutorial for modifying network requests.

> Start by running `mitmproxy`
> Press `i` to open mitmproxy's command prompt, type in `~u /Dunedin & ~q`, and press `Enter`
> Open another terminal, and run `curl --proxy http://127.0.0.1:8080 "http://wttr.in/Dunedin?0"`

Navigating back to our other terminal where we initially ran mitmproxy, we should see a GET request that we have intercepted. Let's go ahead and modify this request.

> Press `Enter` to open the details of the request

Press `e` to edit the request, then select `path` using your arrow keys and press `Enter`
Now press `Enter` again to modify the path component
Replace `Dunedin` with `Innsbruck`, then press `Esc` to confirm the change
Press `q` to return to the details of the request, then press `a` to resume to intercepted request

We should be able to see that the request URL has been updated, and if we navigate back to our other terminal, we should see that `wttr.in` has replied with a weather report from Innsbruck!

🎉 Congratulations, you've completed the lab! 🎉

⚠️ Before moving on, be sure to **turn off the proxy server**, so it doesn't affect your web browser in future assignments!

To do this:

Run `firefox` to open the browser, and click on the hamburger icon in the top right to open the settings
Scroll down to the network settings, and configure Firefox to **NOT** allow a proxy